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Stability analysis of time-delay differential systems with impulsive effect suffered by logic choice

Rong Li [a,1], Zhenhua He [b,*,1]

[a] Office of Academic Affair, Guangxi University of Finance and Economics, Nanning, 530003, PR China
[b] School of Information and Statistics, Guangxi University of Finance and Economics, Nanning, 530003, PR China

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ABSTRACT

In the field of artificial intelligence, nonlinear systems with logic function are everywhere. This paper investigates a time-delay differential system with impulsive effects suffered by logic choice. By the semi-tensor product theory of matrices and constructing Lyapunov function, we analyze the stability of the new system and obtain some stability criteria. The numerical simulations illustrate the validity of these criteria.

1. Introduction

During the past decades, the time-delay differential system attracted a lot of attention since it plays an important role in the fields of science and engineering such as network control systems, population dynamics systems, biological systems. Time-delay systems have been extensively studied and a large number of stability results for time-delay systems were obtained, for example, [1–16] and references therein.

Beside of time-delay phenomena, another an important phenomenon is impulsive phenomena in the process of system evolution. So far, there are many interesting results on impulsive systems. Many scholars studied the properties of this class of impulsive systems, such as the existence, uniqueness and stability of solutions, see, [17–24] and references therein.

It is well known that, however, these classical systems cannot fully characterize the natural phenomena. For instance, the Boolean network [25–27], first introduced by [28], is a powerful tool in describing, analyzing, and simulating the cellular networks. In this model, the state of each gene is determined by the states of its neighborhood genes, using logical rules. Obviously, the Boolean network is a system with logic operation. It does not belong to the common differential system. It is of practical significance to establish a mathematical model based on ordinary differential equations and logic. In some systems biology, the mathematical model is established by ordinary differential equations and some logic rules. For example, the literature [29] pointed out that the fact: Boolean networks are an ideal representation of massive networks that are humble in terms of computational complexity and data demand. However, they are inappropriate when dealing with nested feedback/feedforward loops, structural motifs common in biochemical networks. On the other hand, models of ordinary differential equations (ODEs) cope well with these loops, but they require enormous amounts of quantitative data for a full characterization of the model. Hence, the literature [29] proposed hybrid models, composed of ODE and logical sub-modules, as a strategy to handle large scale, non-linear biochemical networks that include transcriptional and post-transcriptional regulations.

Motivated by the Boolean network model and the classical differential systems, in 2015, the literature [30] introduced a hybrid system-a differential system with impulsive effects suffered by logic choice. This class of systems with logic is complex hybrid

* Corresponding author.
E-mail addresses: rong202103@163.com (R. Li), zhenhuahe@126.com (Z. He).

[1] All authors contributed equally and significantly in writing this paper. All authors read and approved the final manuscript.

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systems, its impulsive effects will change by the logic relation. The authors [30] studied stability of the new system without time-delay and gave some stability criteria. After that, He and Sun [31,32] studied respectively the ultimate boundedness of discrete stochastic time-delay systems and the stability of time-delay discrete systems with logic impulses. To the best of our knowledge, this kind of differential system with logic has not been studied deeply. With the help of theory of semi-tensor product of matrices, in this paper, we will continue to study the time-delay differential system with impulsive effects suffered by logic choice, we obtain some stability criteria for the new system and give some examples to illustrate our results. In the era of artificial intelligence, nonlinear systems with logic function are everywhere. The properties of this kind of system will change with the change of logic function, so it is particularly important to analyze the stability of this kind of system.

The rest of this paper is organized as follows. In the next section, we will introduce the new system. We also give some notations and preliminary results which will be used in what follows. In Section 3, we will give some stability criteria and their proof for the new systems. And, some examples are also given to illustrate the obtained results in this paper. Finally, we will gives some concluding remarks and some prospects for further research.

2. Preliminaries

System description Suppose $\mathbb{R}^m$ and $\mathbb{Z}^+$ denote a set of m dimensional real vectors and a set of positive integers, respectively. In this paper, we consider a class of time-delay differential systems with impulsive effects suffered by logic choice. Such a system is described as follows.

$$\begin{cases} \frac{dx}{dt} = f(t, x, x_t), t \geq t_0, t \neq t_k, \\ x(t_k^+) = x(t_k) := \Phi(x(t_k^-)), \ k \in \mathbb{Z}^+, \end{cases} \tag{1}$$

where $x \in \mathbb{R}^m$, $x_t(\tau) = x(t+\tau), -\xi \leq \tau \leq 0$, ξ is a positive constant, $f(t, x, x_t)$ is continuous for almost all $t \in [t_0 - \xi, +\infty)$ and at the discontinuous points f is right continuous, $f(t, 0, 0) = 0$ for $t \in \mathbb{R}^+$, $x_i(t_k) := f_i I_k(t_k, x_i(t_k^-)) + \overline{f_i} J_k(t_k, x_i(t_k^-))$,

$$f_i = f_i(p_1, \dots, p_m) : \{0,1\}^m \to \{0,1\}$$

is logic functions, $\overline{f_i}$ is the negation logic function of f_i, $p_i = p_i(x_i(t_k^-)) : \mathbb{R} \to \{0,1\}$, $i = 1, \dots, m$, $0 = t_0 < t_1 < t_2 < \cdots < t_k < \cdots$, $t_k \to +\infty$ as $k \to \infty$, $I_k(t, \cdot), J_k(t, \cdot) : [t_0 - \xi, +\infty) \times \mathbb{R} \to \mathbb{R}$ satisfy $I_k(t, 0) = 0, J_k(t, 0) = 0$.

For system (1), we give the following assumptions:

(H1) $|I_k(t, \cdot)| < L_1(t)| \cdot |$, $|J_k(t, \cdot)| < L_2(t)| \cdot |$, $L_1(t), L_2(t) : [t_0 - \xi, +\infty) \to \mathbb{R}^+$.

(H2) The initial conditions associated with (1) are of the form $x_{t_0}(\tau) = x(t_0 + \tau) := \varphi(\tau) \in \mathbb{R}^m, \tau \in [-\xi, 0]$, $\|\varphi\| = \sup_{-\xi \leq \tau \leq 0} \|\varphi(\tau)\|$.

(H3) The system (1) has a unique solution $x(t)$ with the initial condition $x_{t_0} = \varphi$, where the solution $x(t, t_0, \varphi)$ coincides with the function $\varphi(t - t_0)$ for $t \in [t_0 - \xi, t_0]$ and $x(t) : [t_0 - \xi, +\infty) \to \mathbb{R}^m$ is continuous except at the points t_k at which $x(t_k^+) = x(t_k)$, $x(t_k^-)$ exist.

Some notations Let $\Delta_k := \{\delta_k^i | i = 1, 2, \dots, k\}$ and δ_k^i denote the ith column of the identity matrix I_k. $Col_i(B)$ represents the ith column of matrix B, $Col(B)$ denotes the set of columns of matrix B. For two matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$, the semi-tensor product of A and B is $A \ltimes B = (A \otimes I_{\alpha/n})(B \otimes I_{\alpha/p})$, where $\alpha = lcm(n, p)$ denotes the least common multiple and $\otimes$ denotes Kronecker product of matrices; see [33]. Let $\mathscr{K}$ be a set,

$$\mathscr{K} := \{\alpha \in C(\mathbb{R}^+, \mathbb{R}^+) | \alpha(0) = 0, \alpha(s) \text{ is strictly monotone increasing in } s\}.$$

Definition 1 ([33]). A matrix $\mathbb{L} \in \mathbb{R}^{m \times n}$ is called a logical matrix if $Col(\mathbb{L}) \subset \Delta_m$. The set of $m \times n$ logical matrices is denoted by $\mathcal{L}_{m \times n}$.

Lemma 1 ([33]). *Given a logical function $f(p_1, \dots, p_r) \in \Delta_2$ with logical variables $p_1, p_2, \dots, p_r \in \Delta_2$, there exists a unique 2×2^r matrix $M_f \in \mathcal{L}_{m \times n}$, called the structure matrix of f, such that*

$$f(p_1, p_2, \dots, p_r) = M_f \ltimes p_1 \ltimes p_2 \ltimes \cdots \ltimes p_r := M_f \ltimes_{i=1}^r p_i.$$

Moreover, $Col(M_f) \subset \Delta_2$. We note that $\ltimes_{i=1}^r p_i \in \Delta_{2^r}$.

Definition 2 ([17]). Assume $S(\rho) = \{x \in \mathbb{R}^m : \|x\| \leq \rho\}$. The function $V : [t_0, +\infty) \times S(\rho) \to \mathbb{R}^+$ belongs to class v_0 if:

(1) the function V is continuous on each of the sets $[t_{k-1}, t_k) \times S(\rho)$ and for all $t \geq t_0, V(t, 0) \equiv 0$;
(2) $V(t, x)$ is locally Lipschitzian in $x \in S(\rho)$;
(3) for each $k = 1, 2, \dots$, there exist finite limits $\lim_{(t,y) \to (t_k^-, x)} V(t, y) = V(t_k^-, x)$, $\lim_{(t,y) \to (t_k^+, x)} V(t, y) = V(t_k^+, x)$ with $V(t_k^+, x) = V(t_k, x)$ satisfied.

Definition 3 ([17]). Let $V \in v_0$, D^+V is defined as

$$D^+V(t, x(t)) = \limsup_{h \to 0^+} \frac{1}{h} \{V(t + h, x(t + h)) - V(t, x(t))\}.$$

3. Main results

In this section, assume that $V(t) := V(t, x(t)) \in v_0$ is a Lyapunov functional and $V(t^-) := V(t^-, x(t^-))$.

Lemma 2. *Assume $t_k \in [0, +\infty)$, $x_i(t_k) \in \mathbb{R}$ and $x_i(t_k^-)$ exists, $k \in \mathbb{Z}^+, i = 1, 2, \dots, m$. If $x_i(t_k) = f_i I_k(t_k, x_i(t_k^-)) + \overline{f_i} J_k(t_k, x_i(t_k^-))$ satisfies the system (1), then $\|x(t_k)\| \le \sqrt{m}\ell_k \|x(t_k^-)\|$, where $x(t_k) = (x_1(t_k), \dots, x_m(t_k))^T, x(t_k^-) = (x_1(t_k^-), \dots, x_m(t_k^-))^T$, $\ell_k \ge \max\{L_1(t_k), L_2(t_k)\}$, $k \in \mathbb{Z}^+$.*

Proof. Note that f_i is logic functions, $\overline{f_i}$ is the negation logic function of f_i, by Lemma 1, there exists a structure matrix M_i for the logical function f_i such that

$$f_i(p_1, \dots, p_m) = M_i^1 \ltimes_{i=1}^m p_i(x_i(t_k^-)),$$
$$\overline{f_i(p_1, \dots, p_m)} = M_i^2 \ltimes_{i=1}^m p_i(x_i(t_k^-)),$$

where M_i^1 and M_i^2 denote the 1th and 2th row of M_i respectively, $i = 1, \dots, m$.

Assume that $\ltimes_{i=1}^m p_i(x_i(t_k^-)) = \delta_{2^m}^{j_k}$, then for $i = 1, \dots, m$ and $k = 1, 2, \dots$, we have

$$\begin{aligned} x_i(t_k) &= M_i^1 \delta_{2^m}^{j_k} I_k(t_k, x_i(t_k^-)) + M_i^2 \delta_{2^m}^{j_k} J_k(t_k, x_i(t_k^-)) \\ &= (I_k(t_k, x_i(t_k^-)), J_k(t_k, x_i(t_k^-)))Col_{j_k}(M_i). \end{aligned}$$

Without loss of generality, let

$$x_{i_1^k}(t_k) = I_k(t_k, x_{i_1^k}(t_k^-)), \dots, x_{i_{m_1^k}^k}(t_k) = I_k(t_k, x_{i_{m_1^k}^k}(t_k^-)),$$
$$x_{j_1^k}(t_k) = J_k(t_k, x_{j_1^k}(t_k^-)), \dots, x_{j_{m_2^k}^k}(t_k) = J_k(t_k, x_{j_{m_2^k}^k}(t_k^-)),$$

where $\{i_1^k, \dots, i_{m_1^k}^k\} \bigcup \{j_1^k, \dots, j_{m_2^k}^k\} = \{1, \dots, m\}$ and $\{i_1^k, \dots, i_{m_1^k}^k\} \bigcap \{j_1^k, \dots, j_{m_2^k}^k\} = \emptyset$, $m_1^k + m_2^k = m$, $k \in \mathbb{Z}^+$. Then, we have

$$\begin{aligned} \|x(t_k)\|^2 &= x(t_k)^T x(t_k) \\ &= \sum_{s=1}^{m_1^k} I_k^2(t_k, x_{i_s^k}(t_k^-)) + \sum_{r=1}^{m_2^k} J_k^2(t_k, x_{j_r^k}(t_k^-)) \\ &\le m_1^k \max_{1 \le s \le m_1^k} \{I_k^2(t_k, x_{i_s^k}(t_k^-))\} + m_2^k \max_{1 \le r \le m_2^k} \{J_k^2(t_k, x_{i_r^k}(t_k^-))\} \\ &\le m_1^k \max_{1 \le s \le m_1^k} \{L_1^2(t_k)|x_{i_s^k}(t_k^-)|^2\} + m_1^k \max_{1 \le r \le m_2^k} \{L_2^2(t_k)|x_{i_r^k}(t_k^-)|^2\} \\ &\le \|x(t_k^-)\|^2 \left(m_1^k L_1^2(t_k) + m_2^k L_2^2(t_k)\right) \le m\ell_k^2 \|x(t_k^-)\|^2, \end{aligned}$$

that is $\|x(t_k)\| \le \sqrt{m}\ell_k \|x(t_k^-)\|, k = 1, 2, \dots$. We complete the proof.

Theorem 1. *Assume that $D^+V(t) \le 0, t \ne t_k$, $k \in \mathbb{Z}^+$ and there exist two positive constants a, b such that $a\|x(t)\| \le V(t)$ and $V(t_k) \le b\|x(t_k)\|$. If $\{\frac{(\sqrt{m}b)^k \ell_k \cdots \ell_1}{a^k}\}$ is bounded, then the system (1) is stable. Furthermore, if $\lim_{k\to\infty} \frac{(\sqrt{m}b)^k \ell_k \cdots \ell_1}{a^k} = 0$, then the system (1) is asymptotically stable.*

Proof. Since $V(t) \in v_0$, V is locally Lipschitzian in x which implies that there exists a constant $L > 0$ such that $V(t) \le L\|x(t)\|$, $t \in [t_0 - \xi, t_0]$. Assume for $\forall \varepsilon > 0$, $\|\varphi\| < \varepsilon$. For $t \in [t_0 - \xi, t_0]$, note that there exists $s \in [t_0 - \xi, t_0]$ such that $t = t_0 + s$. So, we have

$$V(t) = V(t_0 + s) \le L\|x(t_0 + s)\| < L\varepsilon := \delta(\varepsilon).$$

For $t \in [t_0, t_1)$, by $D^+V(t) \le 0$, we have $V(t) \le V(t_0) \le \delta(\varepsilon)$. Hence, $a\|x(t_1^-)\| \le V(t_1^-) \le \delta(\varepsilon)$, namely, $\|x(t_1^-)\| \le \frac{1}{a}\delta(\varepsilon)$. By Lemma 2 and the condition,

$$V(t_1) \le b\|x(t_1)\| \le \sqrt{m}b\ell_1 \|x(t_1^-)\| \le \frac{\sqrt{m}b\ell_1}{a}\delta(\varepsilon). \tag{2}$$

So, for $t \in [t_1, t_2)$, we have

$$\begin{aligned} V(t) &\le V(t_1) \le \frac{\sqrt{m}b\ell_1}{a}\delta(\varepsilon), \\ \|x(t_2^-)\| &\le \frac{\sqrt{m}b\ell_1}{a^2}\delta(\varepsilon), V(t_2) \le \frac{(\sqrt{m}b)^2 \ell_2 \ell_1}{a^2}\delta(\varepsilon). \end{aligned} \tag{3}$$

For $t \in [t_{k-1}, t_k), k \in \mathbb{Z}^+$, by mathematical induction, we have

$$V(t) \le V(t_{k-1}) \le \frac{(\sqrt{m}b)^{k-1} \ell_{k-1} \cdots \ell_1}{a^{k-1}}\delta(\varepsilon), \tag{4}$$

and

$$V(t_k) \le \frac{(\sqrt{m}b)^k \ell_k \cdots \ell_1}{a^k}\delta(\varepsilon). \tag{5}$$

So, when $\{\frac{(\sqrt{m}b)^k \ell_k \cdots \ell_1}{a^k}\}$ is bounded, from (4)–(5), the system (1) is stable. Furthermore, if $\lim_{k\to\infty} \frac{(\sqrt{m}b)^k \ell_k \cdots \ell_1}{a^k} = 0$, then the system (1) is asymptotically stable. We complete the proof.

Theorem 2. *Let functions $\mu(\cdot) > 0$ and $\frac{1}{q(\cdot)} > 0$ be locally integrable. Let a, b, c, λ be positive constants and $g \in \mathscr{K}$. If the following conditions hold:*

(i) $a\|x(t)\| \le V(t)$ and $V(t_k) \le b\|x(t_k)\|$, $\frac{\sqrt{m}b\ell_k}{a} \le c, k \in \mathbb{Z}^+$,
(ii) for $t \ne t_k, k \in \mathbb{Z}^+$, $D^+V(t) \le \mu(t)q(V(t))$ whenever $V(t+\tau) \le g(V(t)), g(V) > V$ or $V(t+\tau) \le g^-(V(t)), g(V) < V$,
(iii) for $\vartheta > 0$, $g(\vartheta) > \vartheta, \int_\vartheta^{g(\vartheta)} \frac{1}{q(s)}ds \ge \lambda$ (or $g(\vartheta) < \vartheta, \int_\vartheta^{g^-(\vartheta)} \frac{1}{q(s)}ds \ge \lambda$), $\int_{t_k}^{t_{k+1}} \mu(s)ds < \lambda$,

then the system (1) is stable.

Proof. Let ε be an arbitrary positive constant. Suppose $\|\varphi\| < \frac{\varepsilon}{L}$, where L is a constant $L > 0$ such that $V(t) \le L\|x(t)\|$, $t \in [t_0 - \xi, t_0]$. For $t \in [t_0 - \xi, t_0]$, there exists $s \in [-\xi, 0]$,

$$V(t) = V(t_0 + s) = V(t_0 + s) \le L\frac{\varepsilon}{L} = \varepsilon.$$

Take $\delta = \varepsilon$, we spilt two cases to prove.

Case 1. Assume $g(\vartheta) > \vartheta$ for $\vartheta > 0$. Then we can prove $V(t) \le g(\delta)$, $t \in (t_{k-1}, t_k)$, $k \in \mathbb{Z}^+$. If there exists $t_1^* \in (t_0, t_1)$ such that

$$V(t_1^*) > g(\delta) > \delta > V(t_0),$$

then by the continuity of V, we know that there exist $\gamma_{11} \in (t_0, t_1^*)$ and $\gamma_{12} \in (t_0, \gamma_{11})$ respectively such that $V(\gamma_{11}) = g(\delta), V(\gamma_{12}) = \delta$ and

$$V(t) \le g(\delta),\ t \in [t_0 - \xi, \gamma_{11}], \delta \le V(t),\ \forall\, t \in [\gamma_{12}, \gamma_{11}]. \tag{6}$$

By (6), we have

$$V(t+\tau) \le g(\delta) \le g(V(t)),\ \forall\, t \in [\gamma_{12}, \gamma_{11}]. \tag{7}$$

By (7) and the condition (ii), $D^+V(t) \le \mu(t)q(V(t)), t \in [\gamma_{12}, \gamma_{11}]$. By the condition (iii), we have

$$\lambda \le \int_\delta^{g(\delta)} \frac{dV}{q(V)} = \int_{V(\gamma_{12})}^{V(\gamma_{11})} \frac{dV}{q(V)} \le \int_{\gamma_{12}}^{\gamma_{11}} \mu(t)dt \le \int_{t_0}^{t_1} \mu(t)dt < \lambda.$$

This is a contradiction. So, $V(t) \le g(\delta)$, $t \in (t_0, t_1)$. It follows that $V(t_1^-) \le g(\delta)$.

In the interval (t_1, t_2), we can also prove the same conclusion. Otherwise, if there exists $t_2^* \in (t_1, t_2)$ such that $V(t_2^*) > g(\delta) > \delta > V(t_0)$, then by the continuity of V, we know that there exist $\gamma_{21} \in (t_1, t_2^*)$ and $\gamma_{22} \in (t_1, \gamma_{21})$ respectively such that $V(\gamma_{21}) = g(\delta), V(\gamma_{22}) = \delta$ and

$$V(t) \le g(\delta),\ t \ne t_1,\ t \in [t_0 - \xi, \gamma_{21}], \delta \le V(t),\ \forall\, t \in [\gamma_{22}, \gamma_{21}]. \tag{8}$$

By (8), we have

$$V(t+\tau) \le g(\delta) \le g(V(t)),\ \forall\, t \in [\gamma_{22}, \gamma_{21}]. \tag{9}$$

By (9) and the condition (ii), $D^+V(t) \le \mu(t)q(V(t)), t \in [\gamma_{22}, \gamma_{21}]$. By the condition (iii), we have

$$\lambda \le \int_\delta^{g(\delta)} \frac{dV}{q(V)} = \int_{V(\gamma_{22})}^{V(\gamma_{21})} \frac{dV}{q(V)} \le \int_{\gamma_{22}}^{\gamma_{21}} \mu(t)dt \le \int_{t_1}^{t_2} \mu(t)dt < \lambda.$$

This is a contradiction. So, $V(t) \le g(\delta)$, $t \in (t_1, t_2)$. It follows that $V(t_2^-) \le g(\delta)$. Thus, by the mathematics introduce method, we obtain $V(t) \le g(\delta)$, $t \in (t_{k-1}, t_k)$, $V(t_k^-) \le g(\delta)$, $k \in \mathbb{Z}^+$.

On the other hand, by the condition (i) and Lemma 2, we have

$$\begin{aligned} V(t_k) &\le b\|x(t_k)\| \le b\sqrt{m}\ell_k\|x(t_k^-)\| \le \frac{b\sqrt{m}\ell_k}{a}V(t_k^-) \\ &\le \frac{b\sqrt{m}\ell_k}{a}g(\delta) \le cg(\delta). \end{aligned}$$

Let $\delta^* = \frac{1}{a}\max\{g(\delta), cg(\delta)\}$, then $V(t) \le a\delta^*$, namely, $\|x(t)\| \le \delta^*$, which implies that $x(t)$ is stable.

Case 2. Assume $g(\vartheta) < \vartheta$ for $\vartheta > 0$. In this case, it is obvious that $\vartheta < g^-(\vartheta)$ for $\vartheta > 0$. Similar to Case 1, we can prove

$$V(t) \le g^-(\delta) \text{ and } V(t_k^-) \le g^-(\delta), t \ge t_0 - \xi, t \ne t_k, k \in \mathbb{Z}^+.$$

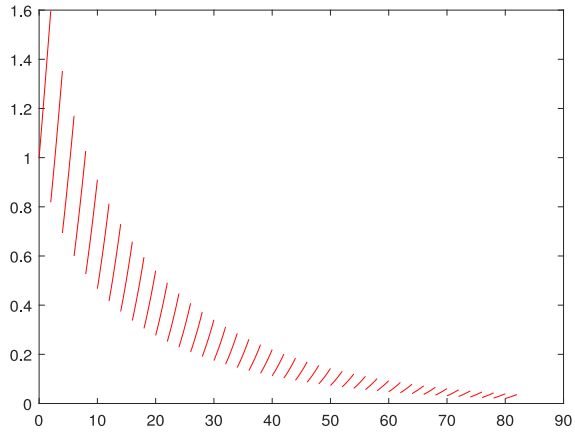

Fig. 1. The state trajectory of system (10).

On the other hand, by the condition (i) and Lemma 2, we have

$$\begin{aligned} V(t_k) &\le b\|x(t_k)\| \le b\sqrt{m}\ell_k\|x(t_k^-)\| \le \frac{b\sqrt{m}\ell_k}{a}V(t_k^-) \\ &\le \frac{b\sqrt{m}\ell_k}{a}g^-(\delta) \le cg^-(\delta). \end{aligned}$$

Let $\bar{\delta} = \frac{1}{a}\max\{g^-(\delta), cg^-(\delta)\}$, then $V(t) \le a\bar{\delta}$, namely, $\|x(t)\| \le \bar{\delta}$, which implies that $x(t)$ is stable. We complete the proof.

Corollary 1. *Let a, b, c, μ, λ be positive constants and $g \in \mathscr{K}$. If the following conditions hold:*

(i) $a\|x(t)\| \le V(t)$ and $V(t_k) \le b\|x(t_k)\|$, $\frac{b\sqrt{m}\ell_k}{a} \le c, \in \mathbb{Z}^+$,

(ii) for $t \ne t_k, k \in \mathbb{Z}^+$, $D^+V(t) \le \mu V(t)$ whenever

$$V(t+\tau) \le g(V(t)), g(V) > V \text{ or } V(t+\tau) \le g^-(V(t)), g(V) < V,$$

(iii) for $\vartheta > 0$, $g(\vartheta) > \vartheta$, $\int_\vartheta^{g(\vartheta)} \frac{1}{s}ds \ge \lambda$ (or $g(\vartheta) < \vartheta$, $\int_\vartheta^{g^-(\vartheta)} \frac{1}{s}ds \ge \lambda$), $\mu(t_{k+1} - t_k) < \lambda$,

then the system (1) is stable.

Proof. Let $\mu(t) \equiv \mu$ and $q(V) = V$ in Theorem 2, then we have Corollary 1.

Numerical examples Let $x(t) \in \mathbb{R}$, $V(t, x(t)) = |x(t)|$, $q(V) = V$, $g(\vartheta) = \frac{1}{3}\vartheta$, $\mu(t) = \frac{1}{2}$, $\lambda = \ln 3$, $t_k = 2k$, $x_0 = 1$, $k = 1, 2, \ldots$.

(I)

$$\begin{cases} \frac{dx}{dt}(t) = \frac{1}{6}\sin x(t) + \frac{1}{6}\sin x(t-3), t \ne t_k, \\ x(t_k) = J_k(t_k, x(t_k^-)) = \frac{1}{1.95}x(t_k^-). \end{cases} \tag{10}$$

It is easy to verify all of the conditions are satisfied in Theorem 2. Therefore, the system (10) is stable. The numerical simulation is shown in Fig. 1.

(II)

$$\begin{cases} \frac{dx}{dt}(t) = \frac{1}{6}\sin x(t) + \frac{1}{6}\sin x(t-3), t \ne t_k, \\ x(t_k) = I_k(t_k, x(t_k^-)) = \frac{1}{3}x(t_k^-). \end{cases} \tag{11}$$

We can verify all of the conditions are satisfied in Theorem 2. The numerical simulation is shown in Fig. 2.

(III)

$$\begin{cases} \frac{dx}{dt}(t) = \frac{1}{6}\sin x(t) + \frac{1}{6}\sin x(t-3), t \ne t_k, \\ x(t_k) = f I_k(t_k, x(t_k^-)) + \bar{f} J_k(t_k, x(t_k^-)), \end{cases} \tag{12}$$

where if $|x(t_k^-)| \ge \frac{2|\sin k|}{k}$, then $f = 1$, otherwise, $f = 0$. We can also verify all of the conditions are satisfied in Theorem 2. Under the threshold conditions $\frac{2|\sin k|}{k}$, $f \not\equiv 0$ or 1, this means that there are two asymptotic behaviors in the system (12). The numerical simulation is shown in Fig. 3.

In Fig. 3, blue line represents the state trajectory under the impulse I_k, red line represents the state trajectory under the impulse J_k. Fig. 3 clearly depicts the trajectory of the system (12) under different impulses.

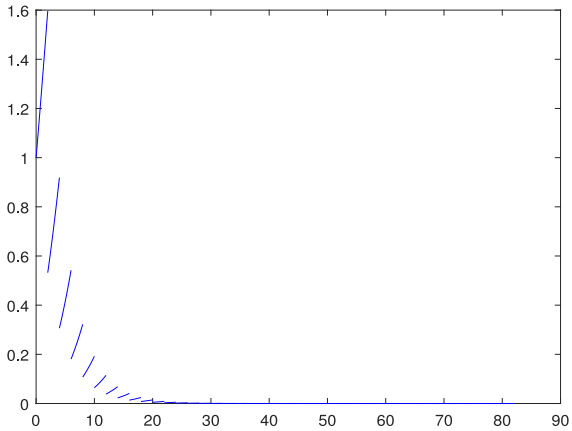

Fig. 2. The state trajectory of system (11).

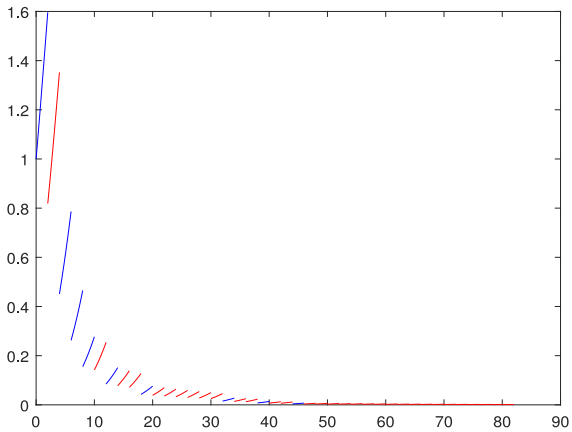

Fig. 3. The state trajectory of system (12).

4. Conclusion

In this paper, we have introduced a new time-delay differential system with impulsive, its impulsive effects are chosen by logic rules. With the help of the semi-tensor product theory, we analyzed stability properties of the new model by constructing Lyapunov functional. We gave numerical examples to illustrate our results.

On the other hand, if $\tau = 0$, then the system (1) will become the system (1) in the literature [30]. By constructing a general Lyapunov functional, our stability criteria are more general than stability criteria in [30].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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